

# HOARDINGS360

## Engineering Handoff — Feature Specification

Scope: V0 (Architecture Foundation) · V1.0 (Owner OS Core) · V1.1 (Field, QR & Proof of Display)

Prepared for: Technical Lead / Engineering Team

*Source of truth: HOARDINGS360 App-wise Feature List by Version (2026-09-17)*

*Document generated: 2026-09-18*

## How to read this document

Every feature from the PRD is written up below as one user story with acceptance criteria, so it can be picked up directly as a backlog item. Scope is fixed at V0, V1.0 and V1.1 only — later versions (V2.0 onward) are out of scope for this document.

**Priority tags** — **CORE** release-blocking, the version does not ship without it. **SHOULD** may slip within the same release if time runs short.

**Module IDs** — M01–M35 map 1:1 to the module IDs in the PRD (e.g. M04 = Compliance). V0 items are grouped under "Foundations" since they precede the module numbering.

**Story IDs** — stable identifiers (e.g. M02-07) for referencing stories in tickets, PRs and standups. Renumber only with product's sign-off.

**Three-status rule** — lifecycle, compliance and occupancy are three independent axes and must always render as three separate badges, never one combined status (see V0-15). This applies across every screen in every version below.

**Cross-cutting conventions** — money in rupee (bigint), timestamps in UTC with business dates in IST, UUID keys plus human board codes, soft delete with a reason everywhere, and role-based field visibility (e.g. floor rate hidden from Sales). See the Foundations section (V0) for the authoritative versions of these rules.

## V0 — Architecture Foundation (Weeks 1–4)

*No customer-facing screens beyond login and an app shell. Everything here is a load-bearing dependency for every later release and cannot be descoped for schedule.*

### Platform Foundations

*Applies to all three apps (Manage, Market, Field) and SuperAdmin unless noted otherwise.*

#### V0-01 Multi-tenancy & row-level security

**CORE** Module: Foundations

**User story:** As a Platform Engineer, I want every business table scoped to a `tenant_id` with PostgreSQL Row-Level Security enforcing isolation, so that one tenant can never read or write another tenant's data, even through a bug.

**Acceptance criteria:**

- `tenant_id` column present and NOT NULL on every business table
- RLS policies enabled on every business table, scoped by `tenant_id`
- An automated cross-tenant test suite runs on every build and blocks the build on failure
- No query path (including admin/reporting queries) bypasses RLS

#### V0-02 Core schema & reserved tables

**CORE** Module: Foundations

**User story:** As a Platform Engineer, I want the full V1.0 data model in place, plus reserved tables for future modules, so that later releases (campaigns, work orders, leases, vendors) can be built without re-architecting the schema.

**Acceptance criteria:**

- All V1.0 entities are modeled: tenants, users, roles, boards, faces, documents, clearances, clients, agreements, occupancy periods, notifications, activity events
- Reserved (empty) tables exist for: campaigns, board\_faces, leases, work\_orders, expenses, vendors
- All schema changes ship through reversible migrations

#### V0-03 Geospatial data layer (PostGIS)

**CORE** Module: Foundations

**User story:** As a Platform Engineer, I want PostGIS-backed location data for every board, with spatial indexing, so that Manage's map, marketplace search and Field's directions can all run fast radius/box/corridor queries.

**Acceptance criteria:**

- A PostGIS point geometry column stores each board's GPS location
- A GiST spatial index exists on the geometry column
- Radius search, bounding-box search and corridor-along-a-route search are all supported at the query layer
- Query performance is validated against a target dataset of 25,000+ boards

#### V0-04 Event bus & async processing

**CORE** Module: Foundations

**User story:** As a Platform Engineer, I want a versioned, immutable event bus with idempotent consumers, so that notifications, audit and the future marketplace projection can react to domain events reliably, including replay after incidents.

**Acceptance criteria:**

- Redis Streams / BullMQ event bus is implemented
- Events are versioned and immutable once published
- Consumers are idempotent — safe to reprocess the same event
- Retry policy and a dead-letter queue exist for failed consumers
- A replay tool can re-deliver events from a given point in time

## V0-05 Status state machines

**CORE** *Module: Foundations*

**User story:** As a Platform Engineer, I want one shared transition service that enforces configured state machines for every status field, so that lifecycle, compliance and occupancy statuses stay consistent everywhere, and frontend code never guesses valid transitions.

**Acceptance criteria:**

- One transition service is used by every status-bearing entity
- Valid states and transitions are configuration, not hardcoded per feature
- Frontend code never compares status strings directly — it reads state via the service/API

## V0-06 Shared entity services: attachments, comments, timeline

**CORE** *Module: Foundations (Manage, Field)*

**User story:** As a Platform Engineer, I want attachments, comments and an activity timeline built once as shared services, so that every entity screen in Manage and Field gets the same behaviour for free instead of being rebuilt per module.

**Acceptance criteria:**

- One attachments service is usable by any entity type
- One comments service is usable by any entity type
- One timeline/activity component is composed into every entity detail screen

## V0-07 Document storage service

**CORE** *Module: Foundations*

**User story:** As a Platform Engineer, I want one polymorphic documents service that every module attaches files to, so that permits, agreements, incidents and every future module store and retrieve files the same way.

**Acceptance criteria:**

- Files are stored in S3 (or equivalent object storage); metadata is stored in the database
- The service is polymorphic — attachable to any entity type
- Delete is soft delete only; there is no hard-delete path

## V0-08 Append-only audit log

**CORE** *Module: Foundations (Manage, SuperAdmin)*

**User story:** As a Platform Engineer, I want an append-only activity\_events table with no update or delete path, so that the audit trail promised to owners is structurally guaranteed, not just a UI convention.

**Acceptance criteria:**

- activity\_events has no UPDATE or DELETE grants/paths at the application or database level
- Every write is an INSERT capturing who, what, when, from-value and to-value
- Used by Manage and SuperAdmin

**V0-09 Tenant configuration layer**

**CORE** *Module: Foundations*

**User story:** As a Platform Engineer, I want alert thresholds, board types, clearance types, incident categories, SLAs and commission % configurable per tenant, with platform defaults, so that every tenant can tailor these lists without a code change, and new tenants get sensible defaults out of the box.

**Acceptance criteria:**

- Configuration entities exist for: alert thresholds, board types, clearance types, incident categories, SLAs, commission %
- Each tenant can override the platform default independently
- Configuration changes take effect without a deploy

**V0-10 Versioned public API**

**CORE** *Module: Foundations*

**User story:** As a Platform Engineer, I want a versioned REST v1 API with a published OpenAPI spec, used by all three client apps, so that Manage, Market and Field share one contract, and the API can evolve without breaking existing clients.

**Acceptance criteria:**

- REST API is versioned under a v1 namespace
- An OpenAPI spec is published and kept in sync with the implementation
- Manage, Market and Field all consume this API — no app-specific backend

**V0-11 Data conventions**

**CORE** *Module: Foundations*

**User story:** As a Platform Engineer, I want consistent conventions for money, time and identifiers across the platform, so that the team avoids rounding bugs, timezone bugs and ID collisions in every future module.

**Acceptance criteria:**

- Monetary values are stored as bigint in Rupee — no floating point currency
- Timestamps are stored as UTC timestampz; business dates are handled in IST
- Primary keys are UUIDs; boards additionally carry a human-readable code (e.g. BLR-HSR-0142)
- Soft delete is used throughout, and every soft delete requires a reason

**V0-12 Environments & CI/CD**

**CORE** *Module: Foundations (Engineering)*

**User story:** As a Platform Engineer, I want local, dev, staging and production environments with CI/CD and reversible migrations, so that releases ship safely and can be rolled back without data loss.

**Acceptance criteria:**

- Four environments are provisioned: local, dev, staging, production
- A CI/CD pipeline runs tests and deploys automatically

- All schema migrations are reversible
- Per-tenant feature flags are supported
- Structured logs and metrics are emitted from all services

### V0-13 Design system

**CORE** *Module: Foundations (UI/UX)*

**User story:** As a Frontend Engineer, I want one component library and design token set shared by Manage, Market and Field, so that the three apps look and behave consistently, screens ship faster, and UI bugs are fixed in one place.

**Acceptance criteria:**

- A component library covers core UI primitives (buttons, forms, tables, badges, modals, etc.)
- Design tokens (colour, spacing, typography) are defined once and consumed by all three apps
- Status badge colours (lifecycle/compliance/occupancy) come from tokens/configuration, never hardcoded per screen

### V0-14 Login & app shell (Manage)

**CORE** *Module: Foundations (Manage)*

**User story:** As a Manage user, I want to log in and see the application shell, even before any other V1.0 module ships, so that V0 proves the auth, navigation and design-system foundations end-to-end before real features are layered on.

**Acceptance criteria:**

- Manage exposes a working login screen against the V0 auth foundation
- Authenticated users land on an app shell (sidebar/top bar/nav) built from the design system
- No business modules (boards, clients, etc.) are required to exist yet

### V0-15 Three-status model (cross-cutting rule)

**CORE** *Module: Foundations*

**User story:** As a Frontend Engineer, I want lifecycle, compliance and occupancy modeled and displayed as three separate status axes everywhere, so that a board that is active, non-compliant and occupied at the same time is never collapsed into one misleading status — the single most important signal for owners.

**Acceptance criteria:**

- Lifecycle, compliance and occupancy are distinct fields/state machines, never merged into one "status"
- Every screen that shows board status renders all three as separate badges
- This rule is enforced from V0 onward, ahead of the modules that populate compliance (M04) and occupancy (M06)

## V1.0 — Owner OS Core, Private Beta (Months 2–4)

*H360 Manage's first real release: modules M01–M10, plus a basic SuperAdmin (M33). This is the V1.0 sales demo — Board 360 and the dashboard matter most.*

### M01 — Identity & Access

#### M01-01 Tenant sign-up

**CORE** Module: M01

**User story:** As a Prospective Owner (CompanyAdmin), I want to sign up my company with a profile, GST number, address, logo and brand colours, so that my tenant is provisioned and branded correctly from day one.

**Acceptance criteria:**

- Sign-up captures company profile, GST number, address, logo upload and brand colour selection
- A new tenant record is created scoped by RLS (see V0-01)
- Logo and brand colours are applied wherever the tenant's brand appears (proof packs, PDFs, etc. in later releases)

#### M01-02 Login & account security

**CORE** Module: M01

**User story:** As a Manage user, I want to log in with email + password, verify my email, reset my password, and confirm privileged role changes with OTP, so that my account and the tenant's data stay secure.

**Acceptance criteria:**

- Email + password login implemented
- Email verification flow on sign-up
- Password reset flow (forgot password)
- OTP challenge required for privileged role changes

#### M01-03 Session tokens

**CORE** Module: M01

**User story:** As a Platform Engineer, I want short-lived JWTs with rotating refresh tokens and server-side revocation, so that compromised sessions can be killed immediately and tokens don't stay valid indefinitely.

**Acceptance criteria:**

- Access tokens are short-lived JWTs
- Refresh tokens rotate on use
- Refresh tokens can be revoked server-side (e.g. on logout-everywhere or admin action)

#### M01-04 Role-based permissions

**CORE** Module: M01

**User story:** As a CompanyAdmin, I want permissions expressed as module.action (e.g. permits.edit, pricing.view\_cost), so that I can grant exactly the access each role needs, no more.

**Acceptance criteria:**

- Permission model is module.action based, not free-text or all-or-nothing

- Every protected action in the API checks the caller's permissions server-side (not just hidden in the UI)
- Default roles match the role table (CompanyAdmin, Operations Manager, Sales Executive, Compliance Officer, etc.)

### M01-05 Invite, deactivate & audit users

**CORE** Module: M01

**User story:** As a CompanyAdmin, I want to invite users with a role, deactivate (never delete) accounts, and see an audit of access changes, so that I always know who has access to what, and can revoke access without losing history.

**Acceptance criteria:**

- Invite-by-email flow assigns a role at invite time
- Users can be deactivated but never hard-deleted
- Every access/role change is written to the audit log (V0-08)

### M01-06 Geography-scoped roles

**SHOULD** Module: M01

**User story:** As an Operations Manager, I want a supervisor role that only sees boards in their own district, so that large, multi-district owners can delegate without over-exposing data.

**Acceptance criteria:**

- Roles can be scoped to a geography (e.g. district)
- A geography-scoped user's queries and lists are filtered server-side to their scope

## M02 — Asset Registry

### M02-01 Board record

**CORE** Module: M02

**User story:** As an Operations Manager, I want to create a board with a system ID, a human-readable code (e.g. BLR-HSR-0142), a name and a configurable structure type, so that every board is uniquely and consistently identifiable across the system and in conversation with the field team.

**Acceptance criteria:**

- Board record stores a system-generated ID and a human-readable code
- Structure type is a configurable list (tenant + platform defaults)
- Board name is a free-text field

### M02-02 Board location

**CORE** Module: M02

**User story:** As an Operations Manager, I want to record a board's GPS pin, address, landmark, city, district, state, PIN, ward, road name and road category, so that the board is findable on a map and in every location-based filter.

**Acceptance criteria:**

- GPS pin captured as a PostGIS point (see V0-03)
- Structured address fields: address, landmark, city, district, state, PIN, ward, road name, road category

- Location fields are required to publish a board

### M02-03 Faces per structure

**CORE** *Module: M02*

**User story:** As an Operations Manager, I want to record multiple faces per structure, each with direction, width × height (auto-calculated sq ft), illumination, visibility and traffic direction, so that two-sided and multi-panel boards are sold, priced and tracked independently.

**Acceptance criteria:**

- A structure can have one or many faces
- Each face independently stores: direction, width, height (sq ft auto-computed), illumination, visibility rating, traffic direction, base rate
- Occupancy, pricing and photos are tracked per face, not per structure

### M02-04 Face commercials

**CORE** *Module: M02*

**User story:** As a Sales Executive / CompanyAdmin, I want to record card rate, a role-restricted floor rate, printing and mounting charges, and rate history per face, so that sales staff can quote confidently without ever seeing the confidential floor rate.

**Acceptance criteria:**

- Card rate is visible to all relevant roles; floor rate is visible only to roles with pricing.view\_cost permission
- Printing and mounting charges recorded per face
- Rate changes are versioned/history-tracked, not overwritten

### M02-05 Board photos

**CORE** *Module: M02*

**User story:** As an Operations Manager, I want to upload day, night and approach photos per face, with capture date shown as "last verified", so that buyers and my own team can trust the board's current appearance.

**Acceptance criteria:**

- Day, night and approach photo slots per face
- Capture date stored and displayed as "last verified"
- At least one photo is required before a face can be published

### M02-06 Board lifecycle

**CORE** *Module: M02*

**User story:** As an Operations Manager, I want to record install date, ownership type (owned / leased-in / managed for third party) and retire a board with a reason, so that the system reflects the true legal and operational state of each structure.

**Acceptance criteria:**

- Install date and ownership type fields on the board
- Retire action requires a reason and moves the board's lifecycle status accordingly (never a hard delete)
- Retired boards remain visible in history and reports

## M02-07 Board 360 page

**CORE** Module: M02

**User story:** As a Manage user, I want one page per board with a header (lifecycle, compliance and occupancy badges), map, faces, gallery, occupancy timeline, compliance, agreements, documents and activity, so that I never have to hunt across screens for everything about one board — this is the sales-demo screen.

### Acceptance criteria:

- Header shows lifecycle, compliance and occupancy as three separate badges (per V0-15)
- Tabs present in V1.0: Overview/map, Faces, Gallery, Occupancy timeline, Compliance, Agreements, Documents, Activity
- Tabs for later-version modules (Incidents, Proof, Marketplace, Costs, Profitability) exist as empty-but-visible states explaining what's coming

## M02-08 Extended site data

**SHOULD** Module: M02

**User story:** As an Operations Manager, I want to optionally add a street-view URL, drone imagery, structural drawings, meter number and a "how to reach" note, so that field staff and buyers get richer context on hard-to-find or complex sites.

### Acceptance criteria:

- Optional fields: street-view URL, drone imagery upload, structural drawing upload, meter number, free-text directions note

## M03 — Document Vault

### M03-01 Upload with virus scan & encryption

**CORE** Module: M03

**User story:** As a Compliance Officer, I want to upload PDF, JPG, PNG, DOCX and XLSX files up to 25 MB, scanned for viruses and encrypted at rest, so that the vault is safe to use for legally sensitive documents.

### Acceptance criteria:

- Accepted formats: PDF, JPG, PNG, DOCX, XLSX; 25 MB max file size
- Every upload is virus-scanned before it is stored
- Files are encrypted at rest

### M03-02 Attach to any entity with a document-type taxonomy

**CORE** Module: M03

**User story:** As a Compliance Officer, I want to attach any document to any entity and classify it by type (licence, NOC, tax receipt, bill, etc.), so that documents are findable by what they are, not just where they were uploaded.

### Acceptance criteria:

- Documents can attach to any entity type via the shared documents service (V0-07)
- Document-type taxonomy is tenant-configurable with platform defaults

### M03-03 Versioning

**CORE** Module: M03

**User story:** As a Compliance Officer, I want a new upload of the same document to supersede the old one, with the full version chain browsable, so that I can always see what changed and when, without losing history.

**Acceptance criteria:**

- Uploading a new version marks the prior version as superseded, not deleted
- Full version chain is viewable from the document detail

## M03-04 Document metadata

**CORE** Module: M03

**User story:** As a Compliance Officer, I want to record authority, reference number, issue and expiry dates, amount, signature status, uploader and checksum for each document, so that compliance and finance can search and verify documents without opening every file.

**Acceptance criteria:**

- Metadata fields: authority, ref no., issue date, expiry date, amount, signature status, uploader, checksum
- Checksum computed automatically on upload

## M03-05 Search & filter

**CORE** Module: M03

**User story:** As a Compliance Officer, I want to search and filter documents by entity, type, authority, expiry window and reference number, so that I can find any document in under a minute.

**Acceptance criteria:**

- Filters available: entity, document type, authority, expiry window, ref no.
- Search returns results across the tenant's full document set

## M03-06 Signed, expiring download links

**CORE** Module: M03

**User story:** As a Compliance Officer, I want documents to be downloadable only via signed, time-limited links, so that documents can't leak through a shared or bookmarked URL.

**Acceptance criteria:**

- Every download URL is signed and expires after a short, configurable window
- Unsigned or expired links are rejected server-side

## M03-07 Bulk upload with auto-classify

**SHOULD** Module: M03

**User story:** As a Compliance Officer, I want to drag-and-drop many files at once, auto-classified by filename, so that onboarding a backlog of paper documents doesn't take all day.

**Acceptance criteria:**

- Drag-and-drop accepts multiple files in one action
- Document type is inferred from filename patterns where possible, editable before confirm

## M04 — Compliance

### M04-01 Clearances per board

#### **CORE** Module: M04

**User story:** As a Compliance Officer, I want to record many clearances per board, each with type, governing body, reference number, dates, renewal cycle, fee and document, so that every licence a board needs is tracked individually, not lumped into one status.

**Acceptance criteria:**

- A board can have many clearance records, each independently typed and dated
- Fields: type, governing body, ref no., issue/expiry dates, renewal cycle, fee, linked document

### **M04-02 Configurable clearance types**

#### **CORE** Module: M04

**User story:** As a CompanyAdmin, I want clearance types to be tenant-configurable with platform defaults, so that each city/owner's actual licence mix (municipal, traffic police, structural, electrical, etc.) is represented accurately.

**Acceptance criteria:**

- Platform ships default clearance types
- Tenant admins can add/edit clearance types for their tenant

### **M04-03 Computed compliance status**

#### **CORE** Module: M04

**User story:** As a Compliance Officer, I want each clearance's status (valid / expiring soon / expired / missing / under renewal) computed automatically and rolled up per board and per tenant, so that I never have to manually track expiry dates in a spreadsheet.

**Acceptance criteria:**

- Status is computed from dates, not manually set
- Status rolls up to a board-level compliance status and a tenant-level compliance summary

### **M04-04 Escalating expiry alerts**

#### **CORE** Module: M04

**User story:** As a Compliance Officer, I want alerts at 90, 60, 30, 15 and 7 days before expiry, then daily after expiry, sent to role-based recipients, so that nothing lapses unnoticed.

**Acceptance criteria:**

- Alert schedule: 90/60/30/15/7 days before expiry, then daily once expired
- Recipients are determined by role configuration (see M07 notification rules)

### **M04-05 Renewal flow**

#### **CORE** Module: M04

**User story:** As a Compliance Officer, I want to mark a clearance under renewal, record the application, upload the new certificate, and have the old record superseded automatically, so that renewals are tracked from application to completion without manual record-keeping.

**Acceptance criteria:**

- "Mark under renewal" action records application date and moves status to "under renewal"
- Uploading the new certificate supersedes the old clearance record and resets status to valid

### **M04-06 Compliance risk dashboard**

**CORE** *Module: M04*

**User story:** As a Compliance Officer, I want a dashboard of expired, expiring-in-30-days and missing-document clearances, with drill-down to the board, so that I can work the riskiest items first instead of scanning every board.

**Acceptance criteria:**

- Dashboard shows counts for expired / expiring in 30 days / missing documents
- Each count drills down to the underlying list of boards/clearances

## M04-07 Auto-unpublish on expired mandatory clearance

**CORE** *Module: M04*

**User story:** As a Compliance Officer, I want a face with an expired mandatory clearance to be automatically taken off sale, blocking quoting unless an admin override is made and audited, so that the owner can never accidentally sell an illegal board.

**Acceptance criteria:**

- Expiry of a mandatory clearance automatically unpublishes the affected face
- Quoting/selling that face is blocked at the API level, not just hidden in the UI
- An admin-only override exists, requires a reason, and is written to the audit log

## M04-08 Per-city renewal checklist templates

**SHOULD** *Module: M04*

**User story:** As a Compliance Officer, I want renewal checklist templates that vary by city, so that I follow the right renewal process for each municipality without memorising local rules.

**Acceptance criteria:**

- Checklist templates configurable per city
- Applied automatically based on the board's city when a renewal is started

## M05 — Clients & Agreements

### M05-01 Client record

**CORE** *Module: M05*

**User story:** As a Sales Executive, I want to record a client's name, GST, billing address, industry, contacts and payment terms, so that I have one place to manage every relationship, not a scattered contact list.

**Acceptance criteria:**

- Client entity with fields: name, GST, billing address, industry, contacts, payment terms

### M05-02 Agreement record

**CORE** *Module: M05*

**User story:** As a Sales Executive, I want to create an agreement referencing a client, dates, value, payment schedule, document and status (campaign optional until V2.1), so that every commercial commitment is tracked formally, not just in an inbox.

**Acceptance criteria:**

- Agreement fields: ref, client, dates, value, payment schedule, document, status
- Campaign field exists but is optional at V1.0 (mandatory linkage arrives with M13 in V2.1)

### M05-03 Agreement line items

**CORE** Module: M05

**User story:** As a Sales Executive, I want to add many faces to one agreement, each with its own rate and dates, so that one contract can cover a multi-board deal with different terms per face.

**Acceptance criteria:**

- Agreement supports multiple line items, each linked to a face with its own rate and date range

### M05-04 Versioned revisions, extensions, terminations

**CORE** Module: M05

**User story:** As a Sales Executive, I want to revise, extend or terminate an agreement with a reason, keeping full version history, so that nothing about a contract's history is ever lost.

**Acceptance criteria:**

- Revisions, extensions and terminations are versioned, not destructive edits
- A reason is required for extension and termination

### M05-05 Agreement renewal alerts

**CORE** Module: M05

**User story:** As a Sales Executive, I want alerts at 90, 60, 30, 15 and 7 days before an agreement ends, sent to the owning sales executive, so that I never let a contract lapse without a renewal conversation.

**Acceptance criteria:**

- Alert schedule matches M04: 90/60/30/15/7 days before end date
- Recipient is the agreement's owning sales executive

### M05-06 Client & agreement search

**CORE** Module: M05

**User story:** As a Sales Executive, I want to search and filter clients/agreements by client, board, city, value and expiry window, so that I can find the deal I'm looking for quickly, even across hundreds of contracts.

**Acceptance criteria:**

- Faceted filters: client, board, city, value range, expiry window

### M05-07 Payment schedule flags

**SHOULD** Module: M05

**User story:** As a Finance-facing Sales Executive, I want expected-vs-received flags on an agreement's payment schedule, so that overdue payments are visible without cross-checking finance records manually.

**Acceptance criteria:**

- Each scheduled payment shows expected vs. received status

### M05-08 Client-facing contract expiry reminders (V1.1)

**CORE** Module: M05 (V1.1)

**User story:** As a Sales Executive, I want the client's own contact to be reminded automatically by email and WhatsApp at configurable days before contract end, toggled on/off per client, so that renewal conversations start earlier and clients aren't surprised when a campaign ends.

**Acceptance criteria:**

- Reminder timing is configurable (days before end)
- Sent via email and WhatsApp to the client's contact
- Can be switched on or off per client

## M06 — Occupancy Engine

### M06-01 Occupancy periods computed nightly

**CORE** *Module: M06*

**User story:** As a Platform Engineer, I want occupancy periods per face recomputed nightly and on every relevant change, so that occupancy state is always accurate without manual recalculation.

**Acceptance criteria:**

- A scheduled nightly job recomputes occupancy periods for every face
- Occupancy also recomputes immediately on any relevant change (new agreement, block, cancellation)

### M06-02 Occupancy states

**CORE** *Module: M06*

**User story:** As a Sales Executive, I want each face to show one of: vacant, on hold, booked future, occupied, becoming vacant, blocked, so that I know at a glance whether a face is sellable right now.

**Acceptance criteria:**

- Occupancy state machine implements exactly these six states (built on V0-05)

### M06-03 Availability query

**CORE** *Module: M06*

**User story:** As a Sales Executive, I want to query which faces are free for a continuous date range, filtered by city and type, so that I can answer "what's free for these dates" instantly instead of checking board by board.

**Acceptance criteria:**

- Query returns only faces free for the entire requested continuous date range
- Filterable by city and structure/board type

### M06-04 Pre-listing window

**CORE** *Module: M06*

**User story:** As a Sales Executive, I want a face to be flagged "Available from <date>" during a configurable pre-listing window (default 30 days, 0–90 configurable) while the current campaign still runs, so that boards sell before they ever go empty — the core value proposition.

**Acceptance criteria:**

- Pre-listing window is configurable per tenant, default 30 days, range 0–90
- Flag appears while the existing campaign is still live

### M06-05 Manual occupancy blocks

**CORE** *Module: M06*

**User story:** As an Operations Manager, I want to manually block a face with a reason (maintenance, owner reserve, structural work), so that faces that can't be sold right now are correctly excluded from availability.

**Acceptance criteria:**

- Manual block requires a reason from a configurable list
- Blocked faces are excluded from availability queries and marketplace (later)

## M06-06 Overlap rejection

**CORE** Module: M06

**User story:** As a Sales Executive, I want the system to reject an overlapping agreement, naming the conflicting one, so that double-booking a face is structurally impossible.

**Acceptance criteria:**

- Attempting to create an agreement/booking that overlaps an existing one is rejected
- The error names the specific conflicting agreement

## M06-07 Vacancy register

**CORE** Module: M06

**User story:** As a Sales Executive / CompanyAdmin, I want a register of days vacant and revenue foregone (at card rate) per face, so that vacancy loss is quantified in rupees, not just felt anecdotally.

**Acceptance criteria:**

- Register computes days vacant per face over a selectable period
- Revenue foregone computed using card rate × vacant days

## M06-08 Upcoming vacancies view

**CORE** Module: M06

**User story:** As a Sales Executive, I want a list of faces going vacant in the next 30/60/90 days with their free-from date, so that sales can start selling a board before the current campaign even ends.

**Acceptance criteria:**

- View lists faces by 30/60/90-day vacancy windows with free-from date

## M06-09 Portfolio availability heat calendar

**SHOULD** Module: M06

**User story:** As a Sales Executive, I want a heat calendar of portfolio availability across the next 180 days, so that I can see supply gluts and gaps across the whole portfolio at a glance.

**Acceptance criteria:**

- Calendar view spans 180 days and visualises availability density across the portfolio

## M07 — Notifications

### M07-01 Notification rule engine

**CORE** Module: M07

**User story:** As a CompanyAdmin, I want to configure rules as event → conditions → recipient roles → channels → template, so that every alert in the system (permits, agreements, incidents, etc.) routes through one configurable engine.

**Acceptance criteria:**

- Rule model: trigger event, conditions, recipient role(s), channel(s), message template
- Used as the routing layer for all other modules' alerts (M04, M05, etc.)

**M07-02 In-app centre & email**

**CORE** *Module: M07*

**User story:** As a Manage user, I want an in-app notification centre and transactional email delivery, so that I see every alert whether or not I'm logged in.

**Acceptance criteria:**

- In-app notification centre lists all notifications for the current user
- Transactional email channel implemented

**M07-03 Preferences & digest**

**CORE** *Module: M07*

**User story:** As a Manage user, I want per-user notification preferences and a choice of immediate, daily or weekly digest, so that I control my own alert volume.

**Acceptance criteria:**

- Users can set channel and frequency (immediate/daily/weekly digest) preferences

**M07-04 Delivery log**

**CORE** *Module: M07*

**User story:** As a CompanyAdmin, I want a delivery log showing status and retries for every notification, visible to tenant admins, so that I can troubleshoot "I never got that alert" complaints.

**Acceptance criteria:**

- Delivery log records status (sent/failed/retried) per notification per recipient
- Visible to tenant admin role

**M07-05 Escalation on unacknowledged critical alerts**

**SHOULD** *Module: M07*

**User story:** As a CompanyAdmin, I want unacknowledged critical alerts to escalate automatically to the next role up, so that urgent issues don't sit unread indefinitely.

**Acceptance criteria:**

- Critical alerts that remain unacknowledged after a configurable window escalate to the next role in the hierarchy

**M07-06 WhatsApp & SMS channels (V1.1)**

**CORE** *Module: M07 (V1.1)*

**User story:** As a Manage user, I want WhatsApp Business API and SMS as notification channels, so that alerts reach me on the channel I actually check first.

**Acceptance criteria:**

- WhatsApp Business API integration implemented
- SMS channel implemented
- Message templates for both channels are submitted for approval in Sprint 1 of V1.1

## M08 — Activity & Audit

### M08-01 Per-entity & tenant-wide activity feed

**CORE** Module: M08

**User story:** As a CompanyAdmin, I want a per-entity timeline and a tenant-wide activity feed, so that I can see the history of one record or the whole tenant's recent activity.

**Acceptance criteria:**

- Every entity detail screen shows its own activity timeline
- A tenant-wide feed aggregates activity across entities

### M08-02 Field-level change history

**CORE** Module: M08

**User story:** As a Compliance Officer / CompanyAdmin, I want field-level change history (who, what, when, from, to, why) for agreements, permits, pricing and specs, so that any dispute about "who changed this rate" is answered instantly.

**Acceptance criteria:**

- Change history captures old value, new value, actor, timestamp and (where provided) reason
- Covers at minimum: agreements, permits, pricing, board specs

### M08-03 Filter & CSV export

**CORE** Module: M08

**User story:** As a CompanyAdmin, I want to filter activity by entity, user, date and event type, and export to CSV, so that I can produce an audit extract for a compliance review without a database query.

**Acceptance criteria:**

- Filters: entity, user, date range, event type
- CSV export of the filtered result set

### M08-04 Retention & legal hold

**SHOULD** Module: M08

**User story:** As a CompanyAdmin, I want retention settings and a legal-hold flag on audit records, so that records subject to a legal matter are never purged.

**Acceptance criteria:**

- Retention period configurable
- Legal-hold flag prevents any purge/retention job from removing flagged records

## M09 — Dashboard

### M09-01 Portfolio summary

**CORE** Module: M09

**User story:** As a CompanyAdmin (Rajesh persona), I want a portfolio summary of structures, faces, occupied, vacant, occupancy % and month-on-month trend, so that I know my business's health the moment I open the app each morning.

**Acceptance criteria:**

- Summary tile set: structures, faces, occupied count, vacant count, occupancy %, MoM trend

## M09-02 Risk strip

**CORE** Module: M09

**User story:** As a CompanyAdmin, I want a risk strip showing permits expired/expiring and boards with no photo or no rate, so that compliance and data-quality risk is visible without opening the compliance module.

**Acceptance criteria:**

- Strip shows counts for: permits expired, permits expiring, missing documents, boards with no photo, boards with no rate

## M09-03 Commercial strip

**CORE** Module: M09

**User story:** As a CompanyAdmin, I want a commercial strip of contracts ending in 30/60/90 days with value at risk, and contracted revenue for this and next month, so that I can see revenue at risk before it's lost.

**Acceptance criteria:**

- Strip shows contracts ending in 30/60/90-day windows with associated value
- Shows contracted revenue for the current and next month

## M09-04 Vacancy loss ranked worst-first

**CORE** Module: M09

**User story:** As a CompanyAdmin, I want vacancy loss ranked worst board first, so that I can act on the biggest leaks immediately, not scroll to find them.

**Acceptance criteria:**

- Vacancy loss list sorted descending by loss amount

## M09-05 Portfolio map (dashboard)

**CORE** Module: M09

**User story:** As a CompanyAdmin, I want a portfolio map coloured by occupancy and compliance, with clustering, so that I can see geographic patterns in risk and vacancy.

**Acceptance criteria:**

- Map pins coloured by occupancy state and by compliance state (togglable)
- Clustering for dense areas

## M09-06 Standard exports

**CORE** Module: M09

**User story:** As a CompanyAdmin, I want to export inventory, occupancy, compliance, contract and vacancy registers as CSV/Excel, so that I can share data with people who aren't in the system.

**Acceptance criteria:**

- Export available for each of: inventory, occupancy, compliance, contract, vacancy registers
- CSV and Excel formats supported

## M09-07 Saved views

**SHOULD** Module: M09

**User story:** As a CompanyAdmin, I want to save a filtered dashboard view for reuse, so that I don't have to reapply the same filters every morning.

**Acceptance criteria:**

- Users can save and reload a named filter/view configuration

## M09-08 Role-wise dashboards

**CORE** *Module: M09*

**User story:** As a Manage user, I want a default home dashboard per role, showing only what that role may see, so that everyone opens the app to the view that matters for their job, with no exposure of data they shouldn't see.

**Acceptance criteria:**

- Each role has a distinct default dashboard
- Dashboard content respects the M01 permission model

## M09-09 Live campaigns view

**CORE** *Module: M09*

**User story:** As a Sales Executive, I want a list of every running agreement with client, faces, start/end dates and days left, so that I always know what's currently live without checking each agreement.

**Acceptance criteria:**

- List shows: client, faces, start date, end date, days remaining, for every currently-running agreement

## M09-10 Revenue dashboard

**CORE** *Module: M09*

**User story:** As a CompanyAdmin, I want contracted revenue broken down by month, board, client and district, so that I understand where revenue is actually coming from.

**Acceptance criteria:**

- Revenue dashboard supports breakdowns by month, board, client and district

## M09-11 Gross revenue per board/face (V1.1)

**CORE** *Module: M09 (V1.1)*

**User story:** As a CompanyAdmin, I want gross revenue per board and per face, computed from agreement values, so that revenue can be judged per asset (net-of-cost profitability arrives later at V4.0).

**Acceptance criteria:**

- Gross revenue computed per board and per face from agreement values
- Explicitly gross — cost netting is out of scope until M22 (V4.0)

## M10 — Bulk Import

### M10-01 Downloadable import template

**CORE** *Module: M10*

**User story:** As an Operations Manager, I want a downloadable template covering boards, faces, clients, agreements and permits, so that I know exactly what data format the importer expects.

**Acceptance criteria:**

- Template covers all five entity types in one guided package

## M10-02 Guided import flow

**CORE** Module: M10

**User story:** As an Operations Manager, I want a guided flow: upload → map columns → validation preview → confirm → summary, so that importing hundreds of boards doesn't require engineering help.

**Acceptance criteria:**

- Flow implements exactly these five steps in order
- Column mapping step lets the user match their sheet's headers to system fields

## M10-03 Import validation

**CORE** Module: M10

**User story:** As an Operations Manager, I want validation of required fields, coordinates (inside India and the stated city), date logic and duplicates (within 25 m or by name), so that bad data is caught before it pollutes the registry.

**Acceptance criteria:**

- Required-field validation
- Coordinate validation: inside India and consistent with the stated city
- Date logic validation (e.g. expiry after issue)
- Duplicate detection: same location within 25 m, or matching name

## M10-04 Partial import with correction file

**CORE** Module: M10

**User story:** As an Operations Manager, I want valid rows to import even when other rows fail, with failed rows returned as a correction file with an error column, so that one bad row never blocks the whole sheet.

**Acceptance criteria:**

- Valid rows commit even if other rows fail
- Failed rows are downloadable as a correction file with a per-row error column

## M10-05 Forgiving input parsing

**CORE** Module: M10

**User story:** As an Operations Manager, I want the importer to accept DD-MM-YYYY, DD/MM/YYYY and Excel date formats; "45k" or "₹45,000" style money; and decimal degrees or a Google Maps link for location, so that I can upload my existing messy Excel sheet as-is.

**Acceptance criteria:**

- Date parser accepts DD-MM-YYYY, DD/MM/YYYY and native Excel date serials
- Money parser accepts shorthand ("45k") and currency-formatted ("₹45,000") input
- Location parser accepts decimal-degree coordinates or a pasted Google Maps link

## M10-06 Bulk photo upload matched by filename

**CORE** Module: M10

**User story:** As an Operations Manager, I want to bulk-upload photos matched to boards by filename, so that hundreds of site photos can be attached without opening each board individually.

**Acceptance criteria:**

- Bulk photo upload matches files to boards using filename convention

**M10-07 Onboarding tracker**

**CORE** *Module: M10*

**User story:** As an Operations Manager, I want a tracker showing % of boards with GPS, photo, rate, permit and agreement complete, so that I know exactly how close the portfolio is to fully onboarded, and what's missing.

**Acceptance criteria:**

- Tracker computes completeness % per data field (GPS, photo, rate, permit, agreement)
- Drill-down to the list of boards missing each field

**Platform — M33 SuperAdmin (basic, V1.0)****PLAT-01 Tenant management**

**CORE** *Module: M33*

**User story:** As a Platform SuperAdmin, I want to create, suspend and configure tenants, assign plans and set per-tenant feature flags, so that the HOARDINGS360 team can operate the multi-tenant platform without database access.

**Acceptance criteria:**

- Create/suspend tenant actions
- Plan assignment per tenant
- Per-tenant feature flags configurable from SuperAdmin

**PLAT-02 Platform health monitoring**

**CORE** *Module: M33*

**User story:** As a Platform SuperAdmin, I want visibility into job queues, event bus lag, failed notifications, error rates and storage, so that the team can catch platform problems before tenants report them.

**Acceptance criteria:**

- Dashboard surfaces: job queue depth, event bus lag, failed notification count, error rate, storage usage

**PLAT-03 Support impersonation**

**CORE** *Module: M33*

**User story:** As a Platform SuperAdmin, I want time-boxed impersonation of a tenant user, requiring a reason and fully audited, so that support can troubleshoot a tenant's issue without a standing backdoor into their account.

**Acceptance criteria:**

- Impersonation sessions are time-boxed and require a stated reason
- Every impersonation session is written to the audit log (V0-08)

## V1.1 — Field, QR & Proof of Display (Month 5)

*H360 Field ships for the first time as an installable, offline-capable PWA. Manage gains incident tracking and proof-of-display review. All items depend on the V0 foundations and V1.0 modules above.*

### M02 additions — Asset Registry

#### M02-09 QR code per structure

**CORE** Module: M02 (V1.1)

**User story:** As an Operations Manager, I want a QR code generated per structure, suitable for a weatherproof sticker, so that field technicians can open the right board instantly on site.

**Acceptance criteria:**

- Each structure has a unique, generatable QR code
- QR encodes a link that opens the board directly in Manage/Field

#### M02-10 Board 360 History tab

**CORE** Module: M02 (V1.1)

**User story:** As a Compliance Officer / CompanyAdmin, I want a History tab on Board 360 combining past and current campaigns with uploaded tax receipts, electricity bills and maintenance records on one timeline, so that the full story of a board is in one place without hunting through documents.

**Acceptance criteria:**

- History tab merges campaign history with uploaded financial/maintenance documents into a single chronological timeline

#### M02-11 Board 360 Incidents & Proof tabs go live

**CORE** Module: M02 (V1.1)

**User story:** As an Operations Manager, I want the Incidents and Proof of display tabs on Board 360 (empty since V1.0) to become functional, so that the placeholder tabs from V1.0 now show real data from M11 and M12.

**Acceptance criteria:**

- Incidents tab lists open/past incidents for the board (from M11)
- Proof tab shows proof-of-display captures per campaign (from M12)

### M11 — Incidents (new)

#### M11-01 Log an incident

**CORE** Module: M11

**User story:** As an Operations Manager / Field Technician, I want to log an incident on a board or face from web or mobile, with mandatory photos, so that problems are captured with evidence at the moment they're found.

**Acceptance criteria:**

- Incident can be created from either Manage (web) or Field (mobile)
- At least one photo is required to submit an incident

#### M11-02 Configurable incident categories

**CORE** *Module: M11*

**User story:** As a CompanyAdmin, I want incident categories configurable per tenant (structural, visibility, electrical, operational, other, as defaults), so that categorisation matches how each owner actually talks about problems.

**Acceptance criteria:**

- Default categories: structural, visibility, electrical, operational, other
- Tenant admins can edit the category list

### M11-03 Incident detail capture

**CORE** *Module: M11*

**User story:** As an Operations Manager, I want to record severity, reporter and description, with the affected campaign auto-linked, so that an incident is immediately traceable to the client campaign it affects.

**Acceptance criteria:**

- Severity, reporter and free-text description captured
- System auto-links the currently-active campaign/agreement on the affected face, if any

### M11-04 Incident lifecycle

**CORE** *Module: M11*

**User story:** As an Operations Manager, I want an incident to move through reported → triaged → assigned → in progress → resolved → verified → closed, with an after-photo required before closing, so that every incident is resolved with evidence, not just marked done.

**Acceptance criteria:**

- State machine implements exactly this seven-state lifecycle (built on V0-05)
- An after-photo is required before an incident can move to closed

### M11-05 Incident dashboard

**CORE** *Module: M11*

**User story:** As an Operations Manager, I want a dashboard of open incidents by severity, ageing, repeat boards and resolution time, so that I can spot chronic problem boards and slow resolution before they become client complaints.

**Acceptance criteria:**

- Dashboard breaks down open incidents by severity and age
- Surfaces boards with repeat incidents
- Shows average/median resolution time

### M11-06 Account manager notification when occupied

**CORE** *Module: M11*

**User story:** As a Sales Executive, I want the account manager to be notified automatically when an incident affects a currently-occupied face, so that I hear about a problem on my client's board before the client does.

**Acceptance criteria:**

- Notification fires via M07 when an incident is logged on an occupied face, routed to the owning account manager

## M11-07 Public incident report link

**SHOULD** *Module: M11*

**User story:** As an Anonymous reporter, I want to report an incident via a QR-linked public page, with no account needed, so that anyone near a damaged board (staff or public) can flag it immediately.

**Acceptance criteria:**

- QR code (M02-09) can route to a public, unauthenticated incident-report form
- Submissions create an incident record scoped to the correct tenant/board

## M12 — Proof of Display (new)

### M12-01 Required evidence set

**CORE** *Module: M12*

**User story:** As an Operations Manager / Field Technician, I want a defined evidence set per campaign — day, night (for lit sites), approach and close-up photos, so that every campaign's proof pack is complete and consistent.

**Acceptance criteria:**

- Evidence set configuration requires day, night (when illumination applies), approach and close-up shots
- Capture is considered incomplete until the required set is met

### M12-02 Geo-check on capture

**CORE** *Module: M12*

**User story:** As a Field Technician, I want captures taken outside a configurable radius (default 150 m) of the board to be flagged for supervisor review, so that proof photos can be trusted to have actually been taken at the board.

**Acceptance criteria:**

- Capture location compared against board GPS at time of upload
- Radius is configurable (default 150 m)
- Out-of-radius captures are flagged and routed to supervisor review, not silently accepted

### M12-03 Branded proof-pack PDF

**CORE** *Module: M12*

**User story:** As an Operations Manager, I want a branded, shareable proof-pack PDF per campaign with photos, coordinates, timestamps and a map, so that I can hand a client indisputable evidence their ad was up, ending start-date disputes.

**Acceptance criteria:**

- PDF includes photos, GPS coordinates, capture timestamps and a map for the campaign
- PDF carries the tenant's branding (logo/colours from M01-01)
- A shareable link to the pack is generated

### M12-04 Periodic re-verification

**CORE** *Module: M12*

**User story:** As a Field Technician, I want periodic re-verification captures to be scheduled during long-running campaigns, so that proof stays current for campaigns that run for months, not just at install.

**Acceptance criteria:**

- System schedules/prompts re-verification captures at intervals during a long campaign

## M32 — Field App (new, PWA)

### M32-01 Field login & device binding

**CORE** *Module: M32*

**User story:** As a Field Technician, I want a long-lived login suited to field use, with device binding, so that I don't have to re-authenticate constantly on site, but a stolen device can be locked out.

**Acceptance criteria:**

- Session lifetime is long relative to Manage's web sessions, tuned for field conditions
- Login is bound to a specific device; binding can be revoked centrally

### M32-02 QR scan to open board

**CORE** *Module: M32*

**User story:** As a Field Technician, I want to scan the QR sticker on a structure and have that board open instantly, so that I never have to search for the right board on a small screen.

**Acceptance criteria:**

- Scanning a valid board QR code (M02-09) opens that board's detail in Field immediately

### M32-03 Today's tasks grouped by area

**CORE** *Module: M32*

**User story:** As a Field Technician, I want today's tasks grouped by area, with a hand-off to the phone's native maps app, so that I can plan my route without re-typing addresses.

**Acceptance criteria:**

- Task list groups today's assignments by geographic area
- Each task can hand off directly to the device's default maps app for navigation

### M32-04 In-app camera with stamping

**CORE** *Module: M32*

**User story:** As a Field Technician, I want photo capture to use the in-app camera only (gallery off by default), stamping GPS, device time, server time and technician identity, so that every photo is provably genuine and attributable.

**Acceptance criteria:**

- Camera capture only — gallery/photo-library upload is disabled by default
- Each photo is stamped with GPS coordinates, device timestamp, server timestamp and the capturing technician's identity

### M32-05 Image compression before upload

**CORE** *Module: M32*

**User story:** As a Field Technician, I want photos to be compressed client-side before upload, targeting under 8 seconds per photo on 4G, so that uploads don't stall on poor site connectivity.

**Acceptance criteria:**

- Client-side compression applied before upload

- Upload time target: under 8 seconds per photo on a 4G connection

### M32-06 Offline capture queue

**CORE** Module: M32

**User story:** As a Field Technician, I want an offline capture queue with auto-sync and clear pending/synced indicators, so that I can keep working at sites with poor or no signal, and trust that everything uploads once I'm back online.

**Acceptance criteria:**

- Captures made offline are queued locally (IndexedDB) and synced automatically when connectivity returns
- UI clearly distinguishes pending (not yet synced) from synced items

### M32-07 Board survey mode

**CORE** Module: M32

**User story:** As a Field Technician, I want a guided survey mode that captures photos, GPS, dimensions and structure type for a board in under 90 seconds, so that onboarding a board in the field is fast enough to do at scale.

**Acceptance criteria:**

- Guided flow captures photos, GPS, dimensions and type in a single session
- Target completion time: under 90 seconds

### M32-08 Report an incident from site

**CORE** Module: M32

**User story:** As a Field Technician, I want to report an incident with photos directly from the site, so that problems get logged the moment they're found, with evidence attached.

**Acceptance criteria:**

- Field app exposes the M11 incident-report flow with photo capture

### M32-09 Guided proof-of-display capture

**CORE** Module: M32 / M12

**User story:** As a Field Technician, I want guided capture of the required evidence set (day, night, approach, close-up) in the Field app, so that I can't accidentally submit an incomplete proof pack.

**Acceptance criteria:**

- Field app walks the technician through each required shot from M12-01
- Out-of-radius captures are flagged per M12-02 for supervisor review

### M32-10 Capture quality checks

**SHOULD** Module: M32 / M12

**User story:** As a Field Technician, I want captures to be checked for blur, darkness and obstruction, with a re-capture prompt, so that bad photos are caught on site, not discovered later by the client.

**Acceptance criteria:**

- Client-side or server-side quality checks flag blurry, dark or obstructed photos
- Technician is prompted to re-capture before the photo is accepted

### M32-11 Low-data mode

**SHOULD** *Module: M32*

**User story:** As a Field Technician, I want a low-data mode for constrained connections, so that the app remains usable on poor or expensive mobile data.

**Acceptance criteria:**

- A low-data mode reduces payload size (e.g. lower-res previews, deferred non-essential syncing)

### M32-12 Regional-language interface

**SHOULD** *Module: M32*

**User story:** As a Field Technician, I want the Field app interface available in regional languages, so that technicians who aren't comfortable in English can use the app confidently.

**Acceptance criteria:**

- Language support planned for Malayalam, Tamil, Kannada, Telugu, Hindi
- Full rollout is assessed at V2.0; V1.1 scope may cover a subset — confirm with product before committing a language list